UNSW Business School/
Information Systems and
Technology Management

GROUP ASSIGNMENT COVER SHEET

GroupID: **M16A-4**

Course: **INFS2822 – Programming of Data Analytics**

Tutorial time: **Monday, 4.00pm-6.00pm**

Assignment Title: **Group Assignment**

Due Date: **21 November 2025**

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Executive Summary

Harbour Brew Market is a growing café chain with multiple stores across Sydney which is well-known for its iced coffee. Recently, management noticed that the iced coffee sales have been inconsistent between different weeks, seasons and store locations. These variations make it difficult for management to make business decisions such as forecasting demand, planning marketing budgets and allocating them effectively.

To address this issue, the Head Office commissioned our analytics team to conduct a visual analysis of the company's historical sales dataset to unpack what factors are driving the changes.

In this project, **sales_units,** which is the number of iced-coffee products sold each week, is used as the main measure of sales performance as the price per bottle is relatively low and stays within a narrow range, so revenue does not change much except when volume changes.

The analysis is structured around three key areas to separate the influence of **external** and **internal** factors on sales outcomes:

1. Difference in sales between holiday and non-holiday weeks. (External)
2. The impact of seasonal and temperature changes on weekly iced-coffee sales. (External)
3. The effectiveness of promotional spending on driving sales. (Internal)

By examining both sides separately, the analysis identifies **what** truly drives iced-coffee demand and **where** Harbour Brew Market resources should be focused on.

Drawing on a series of Data Visualisation Analytical Tools including boxplots, heatmaps, line graphs, bar charts and scatterplots, the report identifies **seasonality and temperature as the strongest demand drivers of sales performance.** Holiday effects contribute only moderately and **promotional spending demonstrates minimal return on investment** even at higher budget levels.

Based on these insights, the report **provides operational and marketing recommendations for demand forecasting, inventory planning, workforce scheduling and the reconsideration of current of promotional strategies** to ensure resources are allocated more effectively across the business.

Problem statement

To identify key factors influencing iced-coffee demand, our analysis addresses three questions:

1. **Holiday effect**
   Are holiday weeks associated with higher sales compared to regular weeks?

2. **Seasonal & Temperature effect**
   How do sales change across different seasons and is this pattern consistent across suburbs?

3. **Marketing Effectiveness**
   Does promotional spending generate meaningful increases in weekly sales?

Data Cleaning

```python
# Step 03 - Basic Data Preparation

# 1. Make sure 'date' is a proper real datetime format
data['date'] = pd.to_datetime(data['date'], dayfirst=True)


# -- Add Useful Columns --


# 2. Create Promo_Flag to seperate 'Promo' and 'Non-promo' weeks
#         - This helps in Question 3 when comparing sales between promotional and non-promotional periods.
data['Promo_Flag'] = data['promo_spend'].apply(lambda x: 'Promo' if x > 0 else 'Non-Promo')

# 3. Create Holiday Label for easier plotting in Question 1
#         - This original dataset only gives a 0 or 1 flag. Adding a label improves readability in charts
data['holiday_label'] = data['holiday_flag'].map({1: 'Holiday Week', 0: 'Non-Holiday Week'})
```

```python
# Step 04 - Basic Data Cleaning and Check


# Ensure data is clean and ready for analysis
print("Number of missing values in each column:")
print(data.isnull().sum())
print("\nDoes the dataset contain any missing values?")
print(data.isnull().any().any())
```

```
Number of missing values in each column:
store_id                  0
store_name                0
suburb                    0
date                      0
year                      0
week                      0
month                     0
season                    0
sales_units               0
price_per_unit            0
sales_revenue             0
avg_temp_c                0
promo_spend               0
holiday_flag              0
median_income             0
education_bachelor_pct    0
population_density        0
Promo_Flag                0
holiday_label             0
dtype: int64

Does the dataset contain any missing values?
False
```

Before analysis, we checked the dataset for issues such as missing values, incorrect formats and inconsistent entries after basic data preparation. **No missing or duplicated records were found and all numerical fields such as sales, temperature, promo_spend) were already clean and usable**. The only preparation needed was converting the date column into a proper datetime format and adding helper fields such as Holiday Label and Promo Flag to support holiday period and promotional analysis.

Data Overview

	sales_units	sales_revenue	avg_temp_c	promo_spend	price_per_unit
count	17600.00000	17600.00000	17600.00000	17600.00000	17600.00000
mean	300.57795	1355.64688	17.96906	885.90909	4.50123
std	71.63791	334.01708	4.39491	2067.01885	0.08646
min	85.00000	383.67000	8.70000	0.00000	4.23000
25%	249.00000	1108.41000	13.80000	0.00000	4.43000
50%	295.00000	1332.36000	17.90000	0.00000	4.50000
75%	348.00000	1581.03000	22.20000	0.00000	4.57000
max	585.00000	2696.97000	27.40000	8000.00000	4.78000

Table 1 – Basic Descriptive Statistic of Overall Dataset

Here is a dataset of **17,600 weekly sales** from Harbour Brew Market stores throughout Sydney. Each table represents sales of a bottled iced coffee brand in the same retail store based on prices, promotions, temperature, and holiday weeks. The dependent variable is sales_units. Related business metrics include **sales_revenue** and **price_per_unit,** while **promo_spend** and **avg_temp_c** are marketing activities and weather events. The **holiday_flag** shows whether a week includes a public holiday.

Descriptive Statistics: The weekly sales average is at around 300 units (sd ~ 72), which ranges at a moderate variance between weeks and between stores. The revenue on average is AUD 1,356, in line with the unit price of AUD 4.50.

Average promotion spend is AUD 886 and ranges from 0–8,000, which suggests intense promotion around campaign peaks rather than a consistent application. The temperature average is 18 °C, and seasonal variations of 8.7 to 27.4 °C are realistic.

The dataset, overall, gives rigorous research avenues to the analysis of the effect of internal drivers (price, promotions) and external drivers within and outside an iced-coffee industry on its sales. **Possible key limitation is that the promotional spending variable does not distinguish between different types of marketing activities, making it difficult to identify which specific promotional strategies are effective or ineffective.**

Question1

In this section, we will use three visualisation charts to analyse the impact of holidays on the sales volume of the Harbour Brew Market.

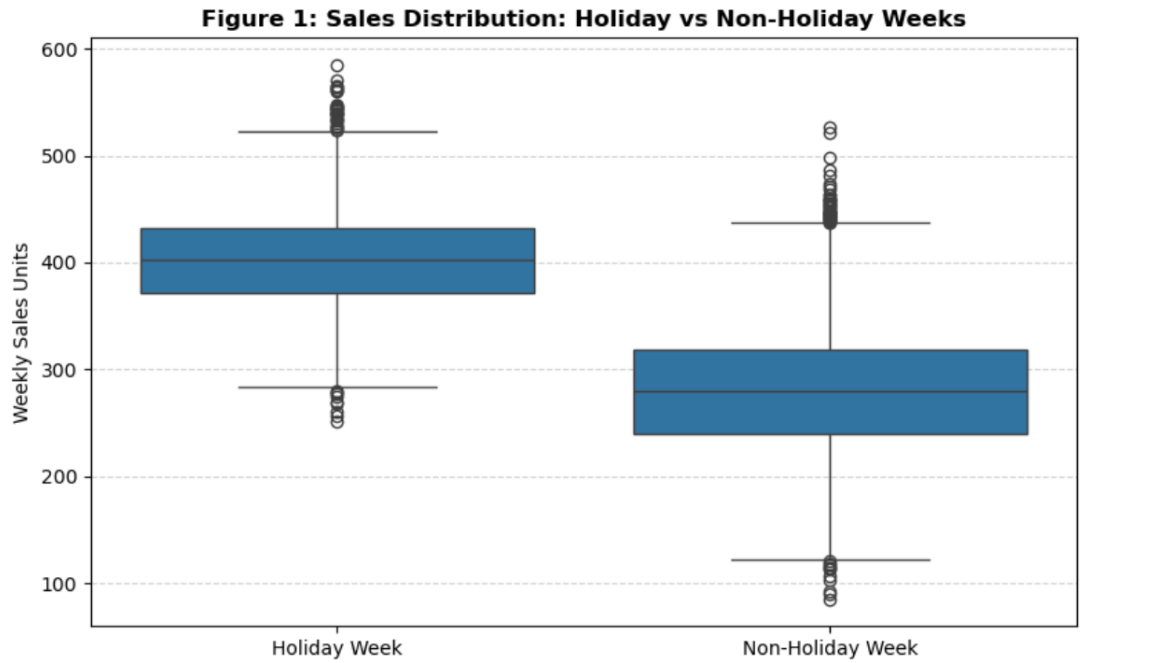
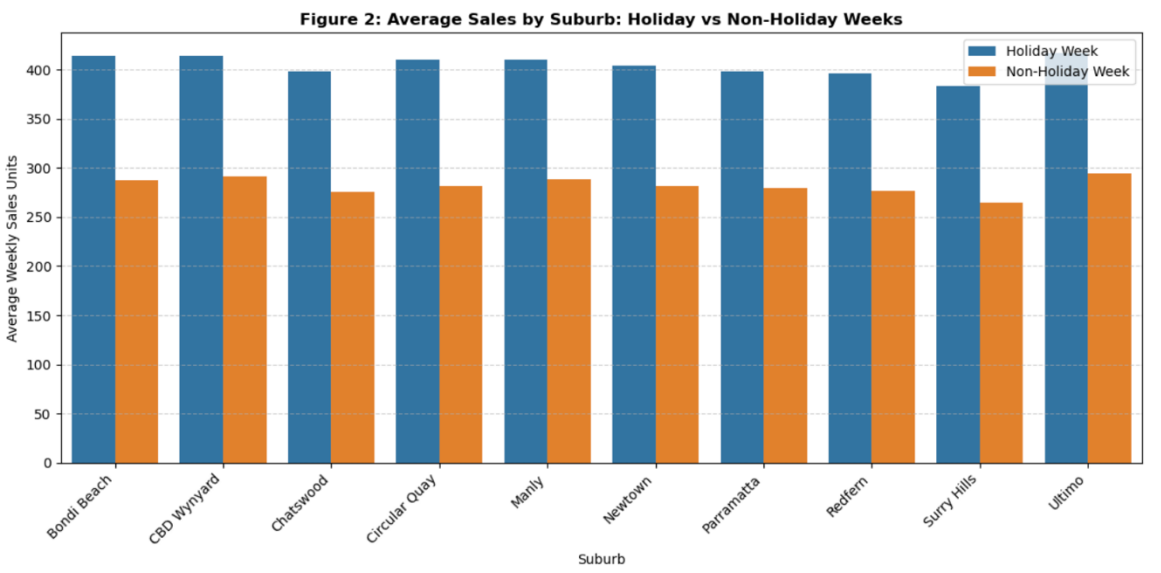
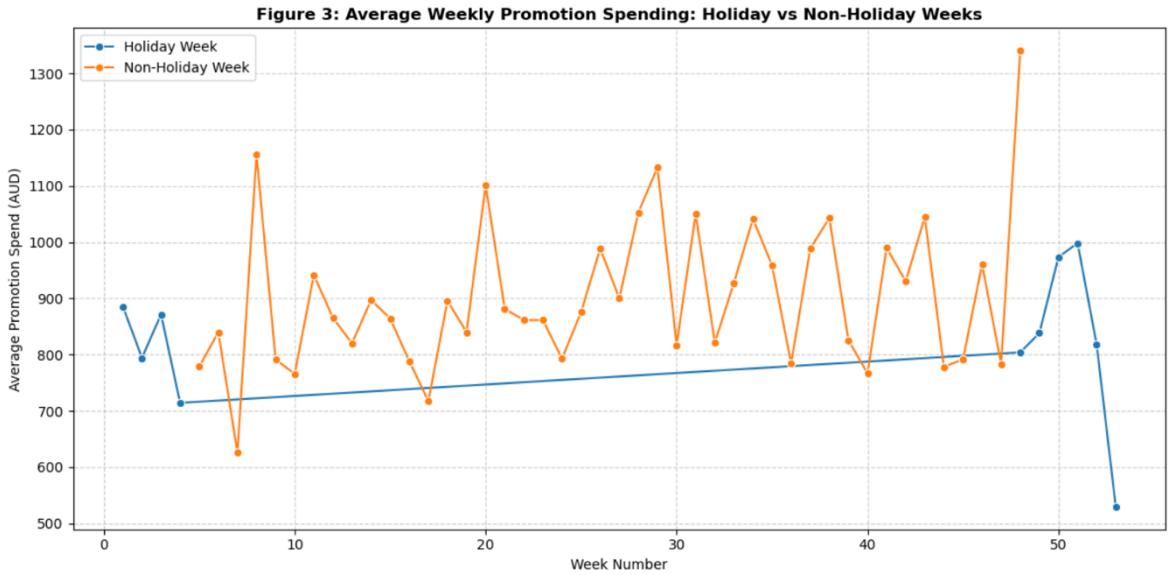


Figure 1 boxplot is used to show the **difference in sales distributions** during holiday and non-holiday weeks. **Holiday weeks clearly show a higher median of average weekly sales** units than non-holiday weeks which means most holiday weeks have stronger demand. Meanwhile, for the interquartile range (IQR), if the IQR of both is relatively high, it indicates that the company's sales volume has a certain degree of volatility. Then, the outliers of the holiday are more concentrated at the top, and a clear overlap can be seen around 550. This instability might be caused by major festivals. On the other hand, outliers during non-holiday periods tend to accumulate at the bottom, indicating that there may be a period of weak daily sales.



In the second part, we delved deeply into the comparison of holiday sales within the region, attempting to determine whether **there are spatial differences in holiday sales**. Figure 2 is a grouped bar chart displays the average weekly sales of each store during holidays and non-holidays. Overall, **the average sales volume during holidays in various regions is higher than during non-holidays.** Furthermore, during holidays, sales in all areas are relatively even, and the sales difference during holidays and non-holidays are also relatively stable. Therefore, it can be considered that **the location of the store has little impact on holiday sales**. Perhaps the focus of the store's marketing should be prioritised on timing rather than locations.



In the third part, it explores whether promotional expenditures have led to an increase in holiday sales. Figure 3 line chart shows the changes in the average weekly promotional expenditure. Overall, **non-holiday promotion spending was slightly higher than holiday promotion spending.** The company's promotional expenditure during non-holiday periods is at a relatively high level and shows multiple peaks, indicating that the enterprise is attempting to continuously use promotional methods to stimulate sales during this period. The fact that holiday promotion expenditures increase significantly only at specific times indicates that they are concentrated on important festivals and the investment is limited. When combined with sales volume, **we can find that marketing has little to do with sales during festivals.** The specific relationship between marketing and sales volume will be analyzed in detail in the third question.

Based on these findings, we put forward the following suggestions for managing holiday-related demand. Firstly, implement flexible inventory and human resource scheduling plans. **Before important festivals, increase the inventory of best-selling products of iced-coffee and temporarily assign additional staff to meet holiday demands** to improve operational efficiency, reduce supply risks, and enhance customer satisfaction. And **reduce inventory during non-holiday periods to control costs**. Secondly, we need to maintain a relatively balanced configuration in each store and shift the marketing focus to **timing**.

season	Holiday_Weeks	Non_Holiday_Weeks
Autumn	0	4600
Spring	0	4100
Summer	2900	1400
Winter	0	4600

Table 2 - Number of Holiday and Non-Holiday Weeks by Season

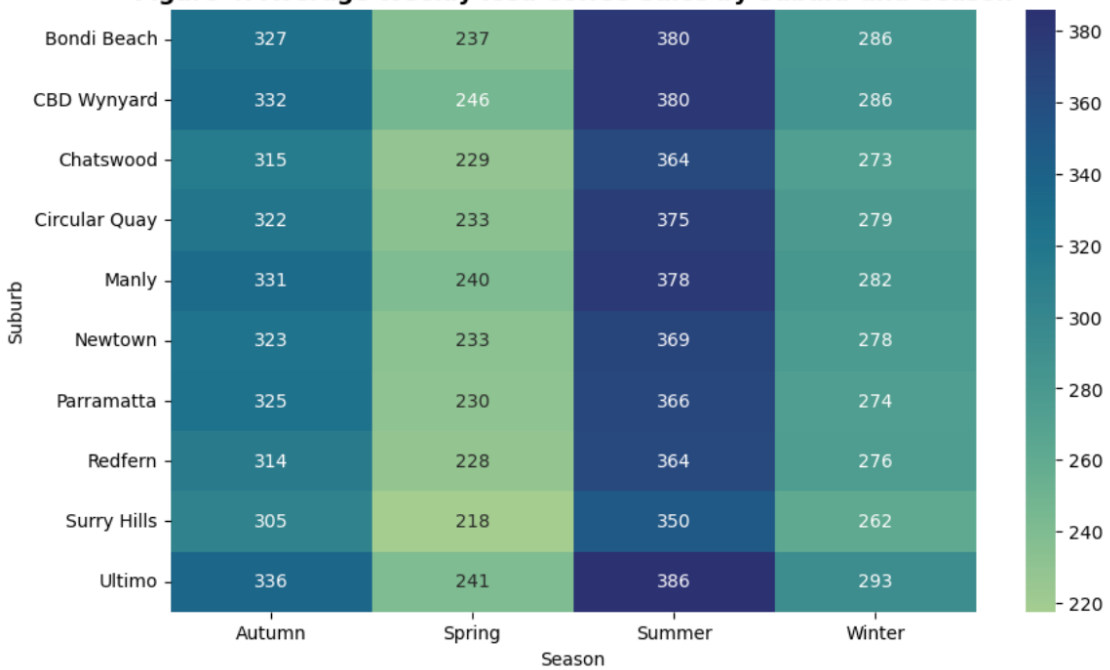
However, one important insight emerged: **All holiday weeks occur during the summer period**. This raises a key question: “Are higher holiday sales truly caused by the holidays themselves, or are they simply the result of seasonal temperature patterns?”

Question2

In the previous section, we found that sales performance during holiday weeks are noticeably higher than during non-holiday periods. According to the dataset, these holidays occurred almost within the summer season. To further explore this relationship, we wanted to confirm whether **the summer season itself consistently records higher sales performance even beyond holiday effect**. We also aimed to identify **which seasons perform better** during the non-holiday weeks.

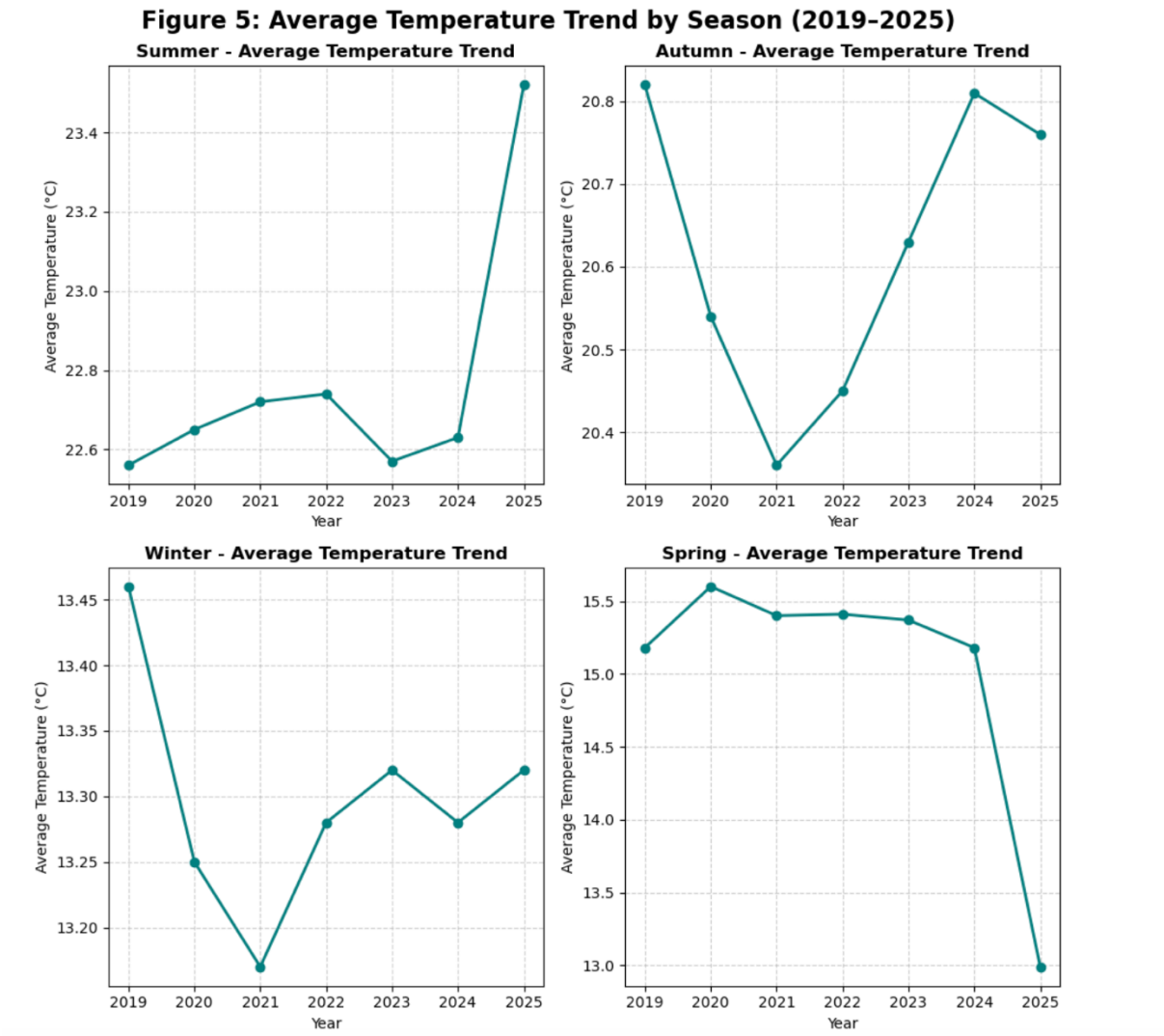
Therefore, we decided to create a heatmap showing the average weekly iced-coffee sales across different Sydney suburbs and seasons. We purposely incorporated suburbs to identify whether there is **any specific location that performs extraordinarily differently** from others during each season.

Figure 4: Average Weekly Iced Coffee Sales by Suburb and Season

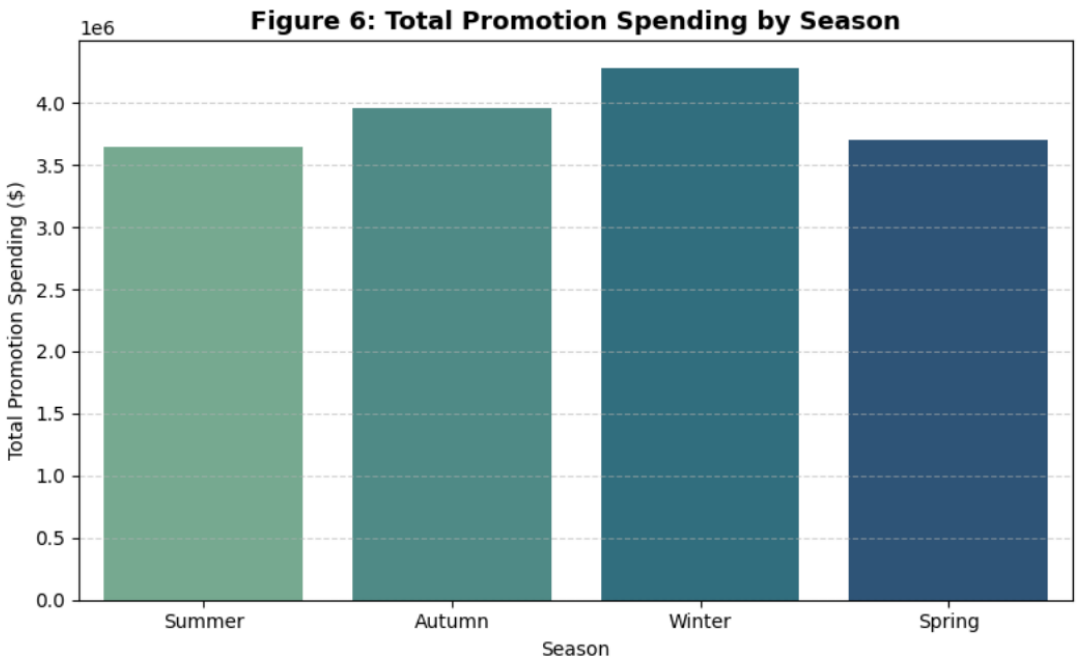


From Figure 4, the heatmap visualises average weekly iced-coffee sales for each suburb across the four seasons(Autumn, Spring, Summer and Winter). The darkest colour represents higher sales volumes.

From the heatmap, it is clear that the **highest average weekly sales units of iced coffee(deepest colour) occur during the summer season** which is around 350-380 bottles. This is expected as customers prefer refreshing and cold drinks to cope with hotter weather. Following summer, **autumn recorded the second-highest sales performance**. To understand this, we compared the average temperature across different seasons.



From Figure 5, we can see that autumn has the second-highest average temperature(20c) after summer(26c). In contrast, winter and spring consistently record the lowest average temperatures across the years. However, one interesting exception appears where **spring shows the lowest average weekly sales while winter performs slightly better.** One possible explanation is that the store are heavily invested on promotion or marketing campaigns during the winter seasons which helped the store maintain slightly stronger sales than expected despite its colder temperature as shown in the graph below (Figure 6).



Under the assumption that winter would likely have performed below spring without that additional promotional support, we can conclude that iced-coffee is strongly season-dependent and its demand increases as temperature rises despite different locations and the store management team can use this nature to help with inventory management and staff control.

Based on these findings, we developed several recommendations:
Firstly, Harbour Brew Market can use this pattern to **forecast weekly demand of iced-coffee** for each season and **adjust the stocking levels and staffing** accordingly. For example, the store can prepare higher iced-coffee stock and roster extra staff especially during Summer and early Autumn particularly for cold-drink preparation. This can help in avoiding stockouts and ensuring they can still maintain fast and good customer service even during the busiest period.

Secondly during colder seasons, the business should **shift attention towards warm beverages instead of relying on iced-coffee promotions**. Based on our previous findings which shows the inverse relationship between temperature and iced-coffee demand, promoting iced-coffee heavily during colder seasons is unlikely to generate strong returns.

Lastly, the company can apply one **unified seasonal strategy across all suburbs**. From the heatmap we generated, every suburb follows the same rise-and-fall seasonal pattern. There is no need for suburb-specific plans. A single operation plan for inventory, staffing and marketing timelines will simplify operation, reduce planning workload and cost, ensuring consistent execution across all stores.

Question3

To evaluate whether promotional spending influences iced-coffee sales, we firstly compared sales performances across promotional and non-promotional weeks.

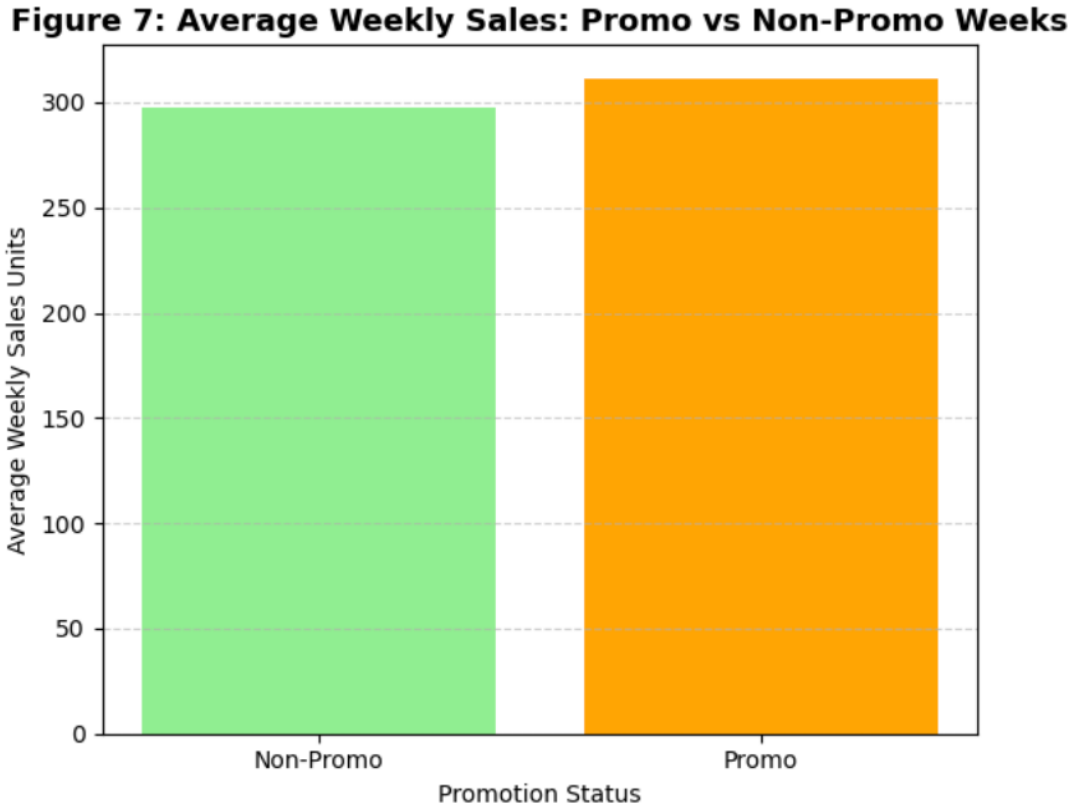
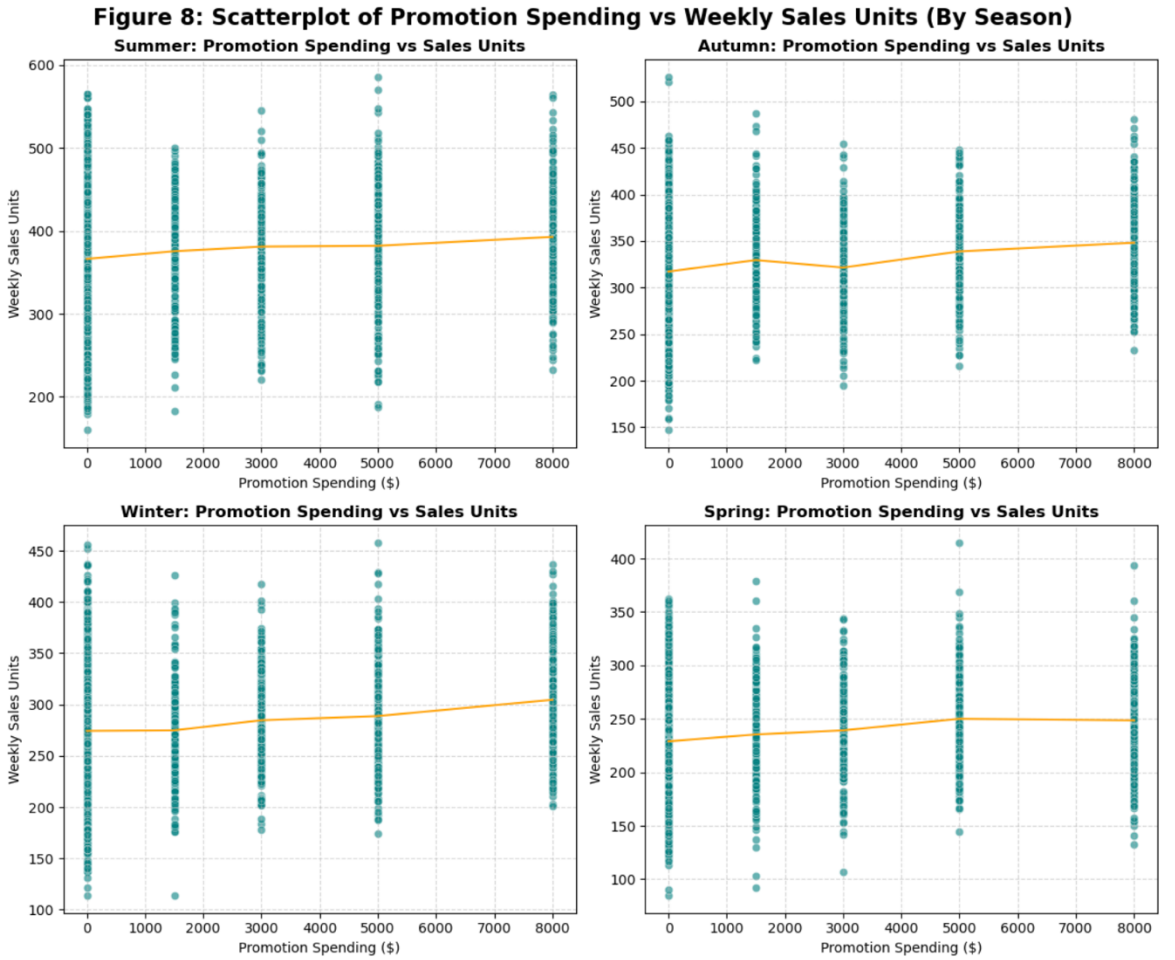


Figure 7 results show that the sales in promotional weeks are slightly higher than non-promotional weeks which indicates that **promotions have limited positive effect on stimulating iced-coffee demand.** Possible reason could be that customers may not be particularly motivated by promotional offers or discounts due to the already low pricing of the iced coffee.

To evaluate the effectiveness of the promotion spending, the analysis will approach the problem from two angles:

Seasonal-level



From Figure 8, the scatterplot shows that the average trend line is almost horizontal which indicates there is no significant correlation between promotional spending and weekly sales performance. **Even as promotional spending increases, the average weekly sales volumes remain relatively unchanged.**

To further confirm the effectiveness of promotional spending on sales performance**,** the analysis focused **solely on promotions week and excluded the non-promotional periods** to avoid dilution of results. The dataset was separated into four seasons to control for environmental influences such as temperature and holidays. The four-line graphs display how average weekly sales units during promotional weeks change with different levels of promotional spending.

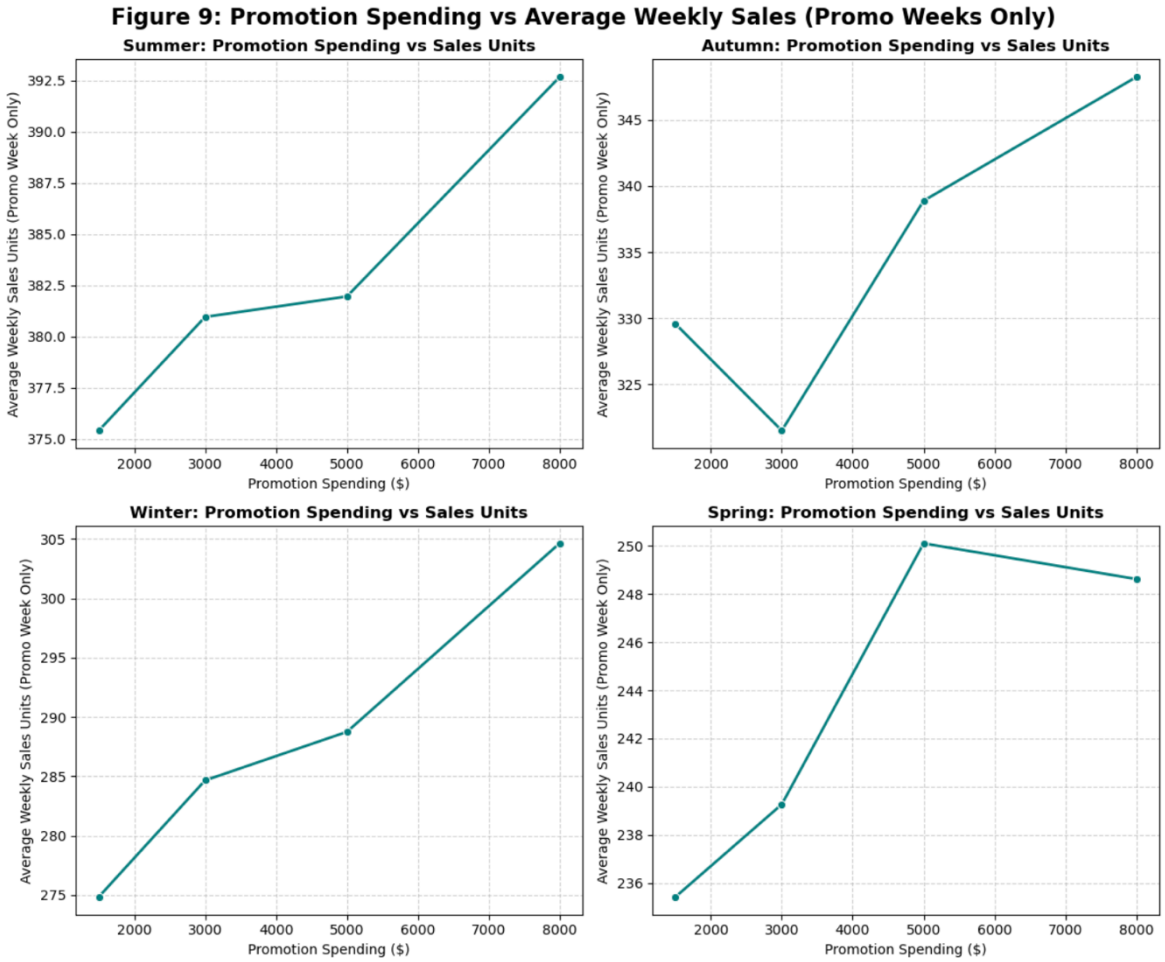
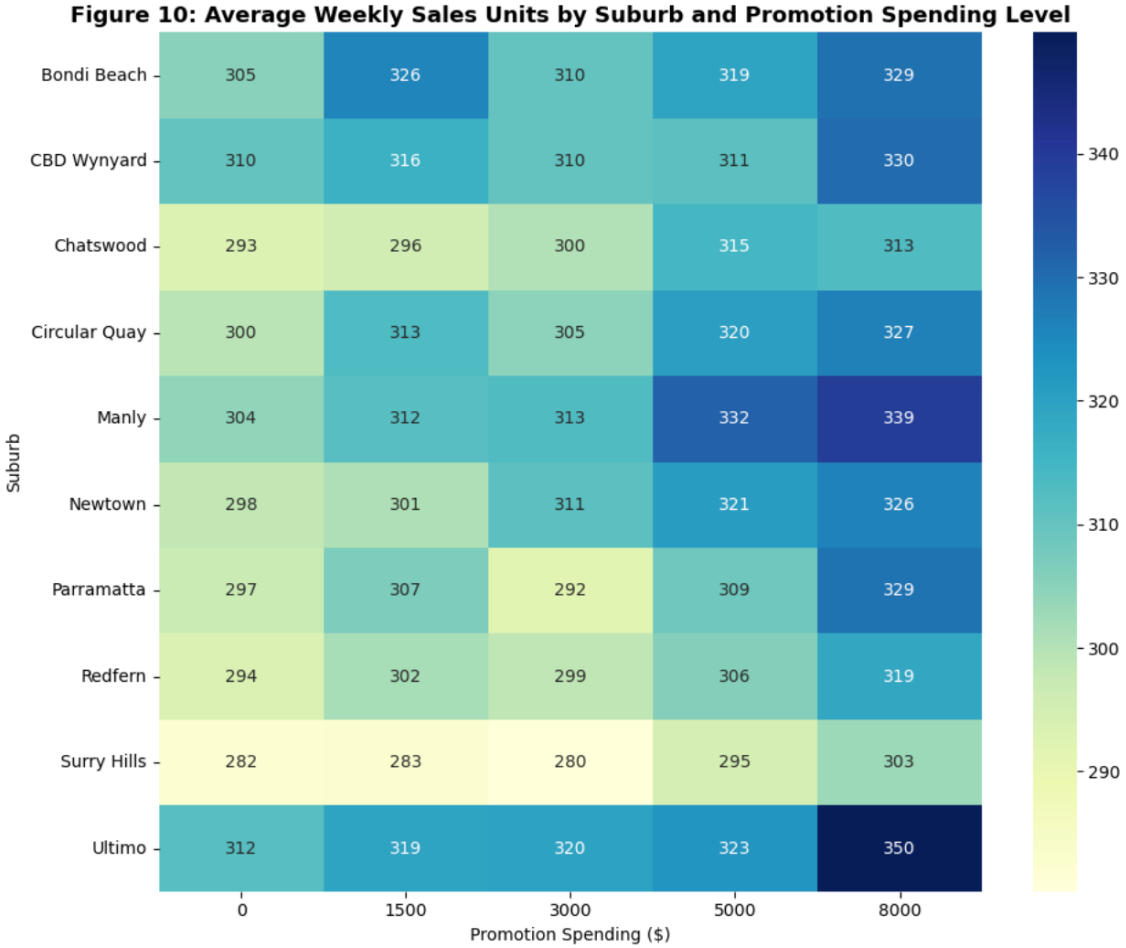


Figure 9 results show that higher spending brings only marginal increases in weekly sales across all seasons. **In winter, where the company invested the most in marketing, average sales rose slightly from around 275 to 305 units as spending increased from $1,500 to $8,000 which indicates an extremely weak return on investment.** A similar pattern appeared in other seasons.

Suburbs-level



To determine whether certain locations might respond better to the promotion, a heatmap(Figure 10) was generated to compare the average weekly sales for different suburbs at different spending levels. While tourist area like Ultimo, Manly and CBD Wynyard show slightly higher overall sales, **no suburb demonstrates strong sensitivity to promotional spending.** For example, increasing promotion spending from $1500 to $8000 lifts suburban sales by only 10-30 units. Given an average iced-coffee price of $4.50, this generates $45-$112 of extra revenue which is far insufficient to justify multi-thousand-dollar weekly promotional budgets. This confirms that promotions remain ineffective even when viewed at a more granular, suburb-specific level.

Based on the findings, recommendations are proposed:

Harbour Brew Market should significantly **reduce iced-coffee promotional spending with the option to eliminate it entirely**. Our previous finding shows that even the highest promotional spendings increased sales by only 10-30 units. Given an average selling price of around $4.50, this produces less than $145 in additional revenue. This means the current promotional activities do not recover even 5% of their cost.

Instead of blindly investing in iced-coffee promotions that do not payoff, Harbour Brew should re-evaluate **the entire marketing expenditure for iced-coffee. This involves**

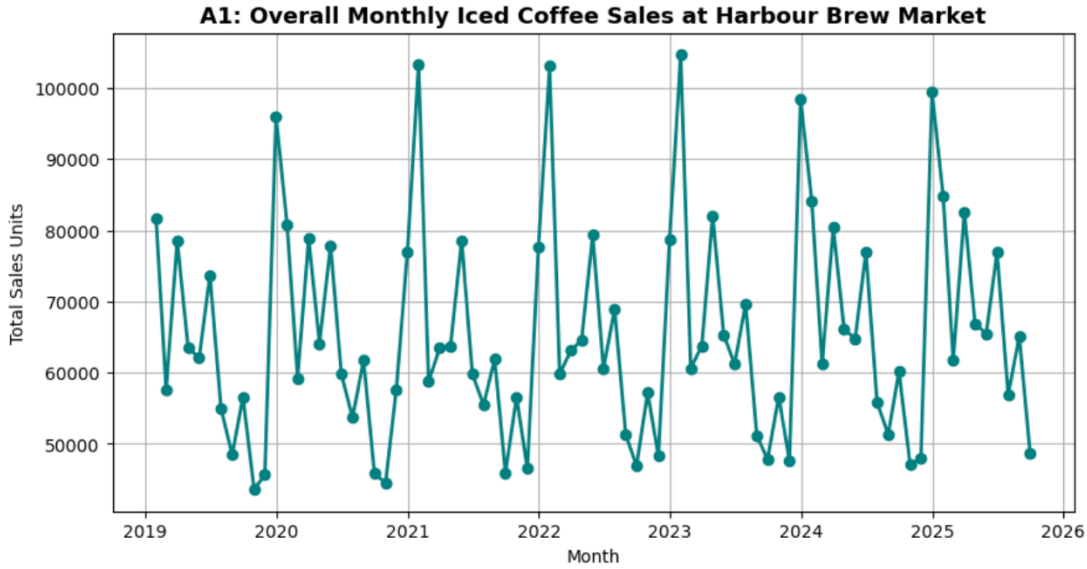
asking: Is the current budget size appropriate or is it unnecessarily high? How exactly is the promotional spending being used and why is it not translating into meaningful sales uplift? Which parts of the promotional activity are ineffective and which can be removed or simplified?

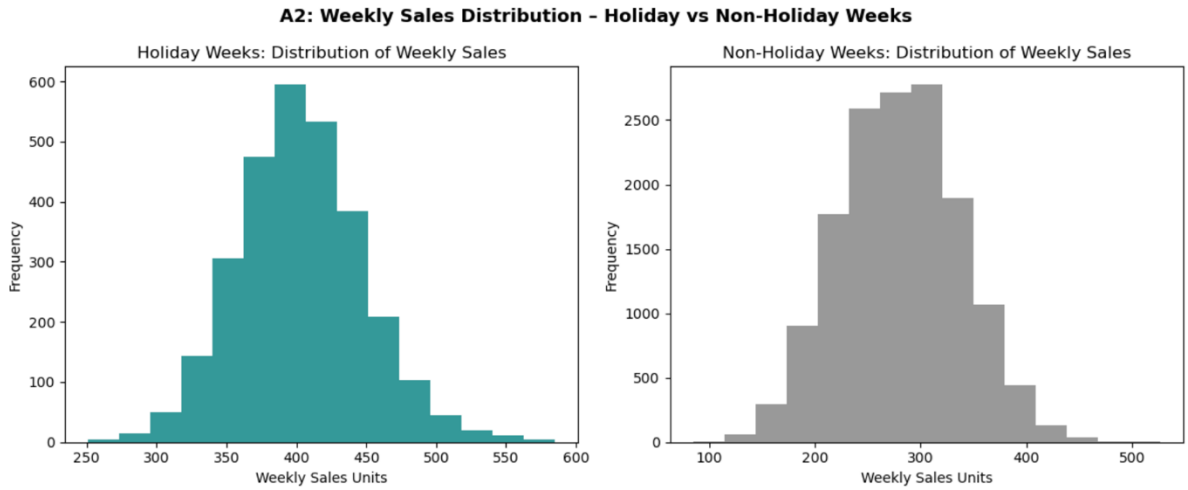
Rather than removing promotions permanently, Harbour Brew should temporarily pause iced-coffee promotional activities and conduct a thorough review. Only after this improvement process should promotional activities be reinstated.

Conclusion

This project examined holiday effects, seasonal patterns and promotional spending to **understand what truly drives iced-coffee sales at Harbour Brew Market**. Across all analyses, the results point to one clear outcome: season and temperature are the dominant factors and remain consistent across all suburbs. Underlying issue identified: High promotional spending shows minimal effectiveness, generating only marginal sales uplift that does not justify its cost. Overall, the store should align operations with seasonal demand and reassess its current promotional strategy to ensure resources are used more effectively.

Appendixes


A1: Overall Monthly Iced Coffee Sales at Harbour Brew Market
Total Sales Units
100000
90000
80000
70000
60000
50000
2019
2020
2021
2022
2023
2024
2025
2026
Month


A2: Weekly Sales Distribution – Holiday vs Non-Holiday Weeks
Holiday Weeks: Distribution of Weekly Sales
Frequency
Weekly Sales Units
Non-Holiday Weeks: Distribution of Weekly Sales
Frequency
Weekly Sales Units